

NETWORK CABLE AND CONNECTORS

1. Draw a labelled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labelling the RJ-45 connector pins.

RJ-45 Connector (Front View, Clip Away)

Pin	1	2	3	4	5	6	7	8
Wire	W/O	O	W/G	B	W/B	G	W/Br	Br

Legend

- W/O = White/Orange
- O = Orange
- W/G = White/Green
- B = Blue
- W/B = White/Blue
- G = Green
- W/Br = White/Brown
- Br = Brown

2. Cable Types and Their Applications

Cable Type	Device/Application	Why Used
<i>Cat5e</i>	WiFi Router	Supports up to 1 Gbps speed and is affordable.
<i>Cat6</i>	Zomato POS Terminal	Faster data transfer and less interference than Cat5e.
<i>Fiber Optic</i>	IPL Streaming Server	Very high speed and long-distance data transmission.

3. Steps to Crimp an RJ-45 Connector

1. Cut the network cable to the required length.
2. Remove about 1 inch of the outer cable jacket.
3. Untwist and arrange the wires in the correct color order (T568B is common).
4. Straighten the wires and trim them evenly.
5. Insert the wires into the RJ-45 connector.
6. Check that all wires reach the end of the connector.
7. Place the connector into the crimping tool.
8. Press the crimping tool firmly to secure the connector.
9. Test the cable using a cable tester.

4. Gaming Zone for IPL Screening Event

Cable: Cat6 Ethernet Cable

Connector: RJ-45 Connector

Reason

- Supports high-speed internet for gaming and IPL streaming.
- Reliable and stable connection with low lag.
- Easy to connect multiple screens, consoles, switches, and routers.

5. Photo of a Network Cable/Connector in Use

Example Description

Type: Cat6 Ethernet Cable with RJ-45 Connector



Location: Behind a Wi-Fi router

Connected Devices: Wi-Fi router and broadband modem

Purpose: Provides internet connectivity from the modem to the router.

Sample Caption

This is a Cat6 Ethernet cable with an RJ-45 connector. It connects a broadband modem to a Wi-Fi router and provides a stable internet connection.